MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 12** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. If PA and PB are tangents to a circle with centre O inclined to each other at $70°$, then $\angle OAB$ is equal to: [1]

(A) $35°$ (B) $45°$ (C) $55°$ (D) $70°$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A) $35°$
Justification: $\angle AOB = 180° - 70° = 110°$. In isosceles $\triangle OAB$ ($OA = OB$): $\angle OAB = \frac{180° - 110°}{2} = \boxed{35°}$.

Q2. In $\triangle ABC$, $AB = 12$ cm, $BC = 16$ cm, and $\angle B = 90°$. A circle is inscribed in the triangle. The inradius of the circle is: [1]

(A) 3 cm (B) 4 cm (C) 5 cm (D) 6 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B) 4 cm
Justification: Hypotenuse $AC = \sqrt{12^2 + 16^2} = 20$ cm. Inradius $r = \frac{12 + 16 - 20}{2} = \boxed{4 \text{ cm}}$.

Q3. If a chord of length 30 cm of a circle of radius 17 cm touches a concentric circle, the radius of the concentric circle is: [1]

(A) 8 cm (B) 10 cm (C) 12 cm (D) 15 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A) 8 cm
Justification: Half-chord = 15 cm. Inner radius $r = \sqrt{17^2 - 15^2} = \sqrt{64} = \boxed{8 \text{ cm}}$.

Q4. A quadrilateral $ABCD$ circumscribes a circle. If $AB = x + 2$, $BC = 6$, $CD = x + 4$, and $DA = 8$, the value of x is: [1]

(A) 2 (B) 3 (C) 4 (D) 5

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) 4

Justification: $AB + CD = BC + DA \implies (x + 2) + (x + 4) = 6 + 8 \implies 2x + 6 = 14 \implies 2x = 8 \implies \boxed{\mathbf{x = 4}}$.

Q5. If tangent segments PA and PB from P to circle with centre O satisfy $PA = 8$ cm and $\angle AOB = 120°$, the length of chord AB is: [1]

(A) 8 cm (B) $8\sqrt{3}$ cm (C) $4\sqrt{3}$ cm (D) 16 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 8 cm

Justification: $\angle APB = 180° - 120° = 60°$. With $PA = PB = 8$ cm, $\triangle PAB$ is equilateral, so $AB = \boxed{\mathbf{8}\text{ cm}}$.

Q6. **Assertion (A):** If tangents PA and PB are drawn to a circle with centre O such that $\angle APB = 90°$, then quadrilateral $OAPB$ is a square.
Reason (R): A quadrilateral having three right angles and adjacent sides equal (radii $OA = OB$) is a square. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $\angle OAP = \angle OBP = \angle APB = 90°$ (fourth angle $\angle AOB = 90°$) and adjacent sides $OA = OB$, directly matching Reason (R) to form a square. Both are true and (R) explains (A).

Q7. If PA and PB are tangents to a circle with centre O from an external point P such that OP equals the diameter of the circle, prove that $\triangle PAB$ is equilateral. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

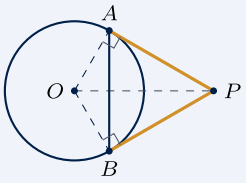
Let radius $OA = r$, so $OP = 2r$.

Step 1: Finding $\angle APB$ [1 Mark]

In right $\triangle OAP$: $\sin(\angle APO) = \frac{OA}{OP} = \frac{r}{2r} = \frac{1}{2} \implies \angle APO = 30^\circ \implies \angle APB = 2(30^\circ) = 60^\circ$.

Step 2: Equilateral Conclusion [1 Mark]

$PA = PB \implies \angle PAB = \angle PBA = \frac{180^\circ - 60^\circ}{2} = 60^\circ$.



$\therefore \triangle PAB$ is equilateral [Proved]

Q8. A quadrilateral $ABCD$ circumscribes a circle. If $AB = 2x + 3$, $BC = 7$, $CD = 3x - 1$, and $DA = 9$, find the value of x and the length of side AB. [2]

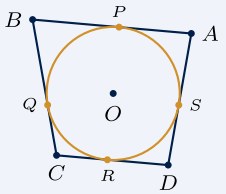
► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Pitot's Opposite-Sum Property [1 Mark]

$AB + CD = BC + DA \implies (2x + 3) + (3x - 1) = 7 + 9 \implies 5x + 2 = 16 \implies 5x = 14 \implies x = 2.8$.

Step 2: Length of Side AB [1 Mark]

$AB = 2(2.8) + 3 = 5.6 + 3 = 8.6$.



$x = \mathbf{2.8}$ (or $\frac{14}{5}$), $AB = \mathbf{8.6}$ units

Q9. A circle touches all four sides of a quadrilateral $ABCD$ whose sides are $AB = x$, $BC = x + 3$, $CD = 2x - 1$, $DA = x + 4$. Find the value of x and the area of the circle if its radius is 3 cm. [3]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

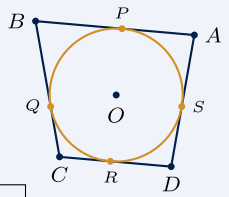
Step 1: Solving for x [1.5 Marks]

For a circumscribed quadrilateral: $AB + CD = BC + DA$

$x + (2x - 1) = (x + 3) + (x + 4) \implies 3x - 1 = 2x + 7 \implies \mathbf{x = 8}$.

Step 2: Area of Circle ($r = 3$ cm) [1.5 Marks]

Area $= \pi r^2 = \pi(3)^2 = 9\pi$ cm^2.

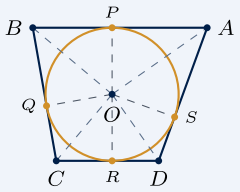


$x = \mathbf{8}$, Area of Circle $= \mathbf{9\pi}$ cm$^\mathbf{2}$ ($\approx$ **28.27** cm$^\mathbf{2}$; **28.26** cm$^\mathbf{2}$ for $\pi = \mathbf{3.14}$)

Q10. Prove that opposite angles of any quadrilateral circumscribing a circle have supplementary central projections, and if the quadrilateral is a trapezium with parallel sides $AB \parallel CD$, then $AB + CD = BC + AD$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let circle (O) touch sides AB, BC, CD, DA at P, Q, R, S.



Part 1: Central Angles Property [1.5 Marks]

By SSS congruence, $\triangle OAP \cong \triangle OAS \implies \angle AOP = \angle AOS$; likewise $\angle BOP = \angle BOQ$, $\angle COQ = \angle COR$, $\angle DOR = \angle DOS$. Sum around O gives $2(\angle AOB + \angle COD) = 360^\circ \implies \angle AOB + \angle COD = 180^\circ$ (and $\angle BOC + \angle AOD = 180^\circ$). [Proved]

Part 2: Opposite-Sides Sum Relation [1.5 Marks]

By Theorem 10.2: $AP = AS$, $BP = BQ$, $CR = CQ$, $DR = DS$. Adding gives $(AP + BP) + (CR + DR) = (BQ + CQ) + (AS + DS) \implies \boxed{\mathbf{AB + CD = BC + AD}}$ [Proved].

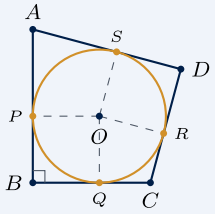
SECTION D: Long Answer / HOTS Type — MATHS MANTHAN | (2 × 5 = 10 Marks)

Q11. A circle with centre O is inscribed in a quadrilateral $ABCD$ touching AB, BC, CD, DA at P, Q, R, S. If $AB \perp BC$, $BC = 38$ cm, $CD = 25$ cm, $AD = 28$ cm, find: (i) the length of side AB, (ii) the radius r of the incircle, (iii) the area of quadrilateral $BPOQ$. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

(i) Length of Side AB: [2 Marks]

Since $ABCD$ circumscribes the circle: $AB + CD = BC + AD \implies AB + 25 = 38 + 28 \implies \boxed{AB = \mathbf{41}\text{ cm}}$.



(ii) Feasibility & Underdetermination Proof for Inradius r: [2 Marks]

Since $\angle B = 90^\circ$ and $OP \perp AB, OQ \perp BC$ with $OP = OQ = r$, $BPOQ$ is a square ($BP = BQ = r$). Tracing tangent segments from B: $CQ = CR = 38 - r \implies DR = DS = 25 - (38 - r) = r - 13 \implies AS = AP = 28 - (r - 13) = 41 - r$. Thus $AB = (41 - r) + r = 41$, where r **cancels identically** (an extra datum such as $\angle D = 90^\circ$ or BP is required to fix r).

Furthermore, checking the Triangle Inequality: with $\angle B = 90^\circ, AB = 41$ cm, $BC = 38$ cm, diagonal $AC = \sqrt{41^2 + 38^2} = \sqrt{3125} \approx 55.90$ cm $> AD + CD = 28 + 25 = 53$ cm. Hence these printed measurements admit **no valid Euclidean quadrilateral and no real numerical inradius** r (typographical omission/error in test paper).

(iii) Area of Quadrilateral $BPOQ$: [1 Mark]

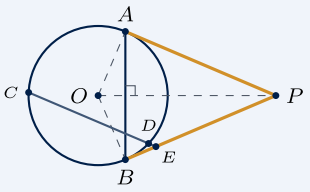
Since $BPOQ$ is a square of side r, its area is $\boxed{\text{Area}(BPOQ) = \mathbf{r^2}\text{ cm}^2}$ *(indeterminate numerically as shown above; diagram is schematic).*

Q12. Two tangents PA and PB are drawn to a circle with centre O from an external point P. A chord CD is drawn parallel to tangent PA such that it intersects PB at E and the circle at C and D. Prove that: (i) OP is perpendicular to AB, (ii) $\triangle PAB$ is isosceles, (iii) If $\angle APB = 40°$, find all interior angles of $\triangle PAB$ and the central angle $\angle AOB$. [5]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

(i) Proving $OP \perp AB$: [1.5 Marks]

Let OP intersect AB at M. In $\triangle APM$ and $\triangle BPM$: $PA = PB$ (Theorem 10.2), $PM = PM$, $\angle APM = \angle BPM$ ($\because \triangle OAP \cong \triangle OBP$). By SAS congruence, $\triangle APM \cong \triangle BPM \implies \angle AMP = \angle BMP = 90° \implies OP \perp AB$. [Proved]



(ii) Proving $\triangle PAB$ is Isosceles: [1.5 Marks]

Since tangents from an external point to a circle are equal in length (Theorem 10.2), $PA = PB \implies \triangle PAB$ is isosceles. [Proved]

(iii) Interior Angles of $\triangle PAB$ and Central Angle $\angle AOB$ (given $\angle APB = 40°$): [2 Marks]

In isosceles $\triangle PAB$: $\angle PAB = \angle PBA = \frac{180° - 40°}{2} = 70°$. In quad $OAPB$: $\angle AOB = 180° - 40° = 140°$.

$$\boxed{\text{Angles of } \triangle PAB : \mathbf{40°},\ \mathbf{70°},\ \mathbf{70°};\quad \angle AOB = \mathbf{140°}}$$

SECTION E: Case-Based / Applied Situation — MATHS MANTHAN | (1 × 4 = 4 Marks)

Q13. Circular Archery Target with Concentric Scoring Rings

An Olympic archery target consists of concentric scoring circles. The inner gold ring has radius $r = 6$ cm and the adjacent red ring has outer radius $R = 10$ cm, both centred at O. An arrow path marks a straight chord AB on the red ring that touches the boundary of the gold ring at M.

(i) State why triangle $\triangle OMA$ is a right triangle. [1]

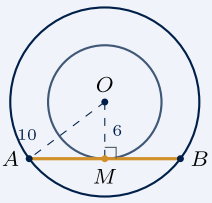
(ii) Find the length of half-chord AM. [1]

(iii) Calculate the full length of chord AB, OR find the area of the red ring between the two concentric circles. (Use $\pi = 3.14$) [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Why $\triangle OMA$ is a Right Triangle: [1 Mark]

Chord AB touches the inner circle at M; by Theorem 10.1, radius $OM \perp AB \implies \boxed{\angle \mathbf{OMA = 90°}}$.



(ii) Length of Half-Chord AM: [1 Mark]

In right $\triangle OMA$: $AM = \sqrt{OA^2 - OM^2} = \sqrt{10^2 - 6^2} = \sqrt{64} = \boxed{\mathbf{8}\text{ cm}}$.

(iii) Main Choice — Full Length of Chord AB: [2 Marks]

Perpendicular OM bisects chord AB, so $AB = 2 \times AM = 2 \times 8 = \boxed{\mathbf{16}\text{ cm}}$.

— OR — (Alternative Choice — Area of Red Ring Between Concentric Circles): [2 Marks]

Area $= \pi(R^2 - r^2) = 3.14(10^2 - 6^2) = 3.14(100 - 36) = 3.14 \times 64 = \boxed{\mathbf{200.96}\text{ cm}^2}$.

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